Pre−Pandemic Data Only

antiGen
ESM 2 (35M)
BLOSUM62
EVEscape
ESM 2 (650M)

antiGen Finetunes

antiGen
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) partial FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as EVE full FT

Pandemic Data in Training Window

antiGen
PSSM Conservation
MSA Trans. (BA.2 eval)
Tranception
PSSM
MSA Transfomer
EVE
EVmutation

Deep Mutational Scans

antiGen
BA.1 DMS (expression)
BA.2 DMS (expression)
BA.1 DMS (bind)
BA.2 DMS (bind)

all mutations, 1+ cases

de novo, 1+ cases

all mutations, 100+ cases

de novo, 100+ cases

A B C D E F G H I J K L M N O P

P(mut. in top decile obs. by t)

Date (t, years into pandemic)